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Causal Inference in Networks (Part 3)

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Outline

Introduction

No Interference

- Outcome of one individual assumed to be unaffected by the treatment assignment of others
- Typical assumption of causal inference
- Part of SUTVA

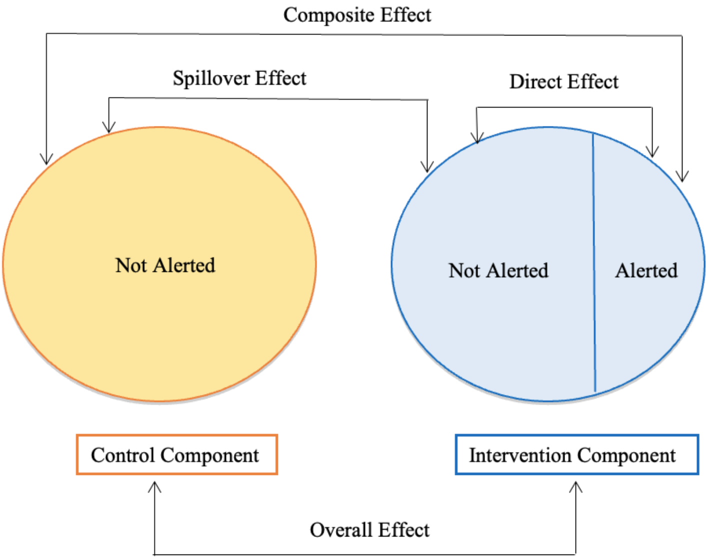
Clearly not true in some settings

- Infectious diseases, education interventions, social sciences
- Individuals often embedded in networks even if we ignore this in our study

Phenomenon of interest vs. nuisance

Halloran and Struchiner (1991, 1995)

- The following slide shows different possible vaccine effects described by HS
- Several vaccine studies have been conducted or analyzed with the intent to estimate these effects (Moulton et al 2001; Longini et al 2002; Ali et al 2005; King et al 2006)
- Overlap in nomenclature with mediation literature (direct, indirect)


Composite Effect
Spillover Effect
Direct Effect
Not Alerted
Not Alerted
Alerted
Control Component
Intervention Component
Overall Effect

Interference Definition

- That two potential outcomes sufficiently represent all potential outcomes for an individual assumes no interference between individuals
  i.e., the treatment of one individual does not affect the outcome of other individuals (Cox 1958)
- The no interference assumption may not hold in some settings
- Examples: Vaccine studies, educational intervention studies, HIV prevention studies
- Settings: Epidemiology, medical research, econometrics, social networks

SUTVA (HR Fine Points 1.1-1.2)

No interference is part of SUTVA (Rubin 1980)
Stable Unit Treatment Value Assumption

1. No interference
2. Only one version of treatment and one version of no treatment (control)

Or if there are multiple versions of treatment, they are irrelevant (Cole and Frangakis 2008, VanderWeele 2009, Pearl 2010)

See HR Fine Point 1.2 regarding multiple versions of treatment

General Approach

Population of groups of individuals (blocks of units; clusters)
Assume partial interference: Possible interference between individuals in a group but not between groups.
Define direct (individual), disseminated (indirect), composite (total), overall causal effects
Two-stage randomization

1. Groups to allocation strategies α_1, α_0
2. Given 1, individuals randomized to treatment/controls $A \in 0, 1$

Unbiased estimators, variance using randomization-based inference or M-estimation (Hudgens and Halloran, 2008)

Example: Vaccine Trial

Groups: Schools sufficiently separated geographically
Individuals: Students
Assignment mechanism

1. Randomized some schools to 50%, others to 25% vaccine coverage
2. Randomized students to vaccine or placebo conditional on school assignment strategy from step 1

Notation (H&H 2008)

- N groups; n_k individuals in groups $k = 1, \ldots, N$
- $\mathbf{A}_i = (A_{1i}, \ldots, A_{1n_i})$ treatments received for n_i individuals in group i
  $A_{ij} = 0$ or 1 implies $\mathbf{A}_i$ can take on 2^{n_i} possible values
  $\mathbf{A}_{i,-j}$ is the $n_i - 1$ subvector of $\mathbf{A}_i$ with the j^{th} entry deleted
  $\mathbf{a}_i$ and a_{ij} denote possible values of $\mathbf{A}_i$ and A_{ij}
- Let $A(n)$ be the set of vectors of all possible exposure allocations of length n. e.g., $A(2) = \{(0,0),(0,1),(1,0),(1,1)\}$, $\mathbf{a}_i \in R^{n_i}$
- $A(n,k)$ denotes when exactly k individuals receive treatment 1 (i.e., completely randomized design)
- Let α be the proportion assigned to treatment in a group

Assignment Mechanism

- $S_i = 1$ if the i^{th} group is assigned to α_1 and 0 otherwise
  $\mathbf{S} = (S_1, \ldots, S_N)$
  $C = \sum_i S_i$
- Parameterization for treatment assignment strategy
  - Complete randomized group assignment strategy if k_i number treated in block i , i.e., $\pi(a_i, \alpha) = I(a_i \in A(n_i, k_i))/\binom{n_i}{k_i}$
  - Bernoulli allocation: $\pi(\mathbf{a}_i, \alpha) = \prod_{j=1}^{n_i} \alpha^{a_{ij}}(1-\alpha)^{1-a_{ij}}$

Potential Outcomes

- $y_{ij}(\mathbf{a}_i)$ is the potential outcome of individual j in group i under $\mathbf{a}_i$
- Allows interference between individuals within group i
- Can write $y_{ij}(\mathbf{a}_i)$ as $y_{ij}(\mathbf{a}_{i,-j}, a_{ij} = a)$
- Have 2^{n_i} potential outcomes per individual, instead of 2 potential outcomes per individual in the absence of interference

Average Potential Outcomes

Individual average potential outcome

$$\bar{y}_{ij}(a, \alpha) = \sum_{a_{i,-j} \in A(n_i - 1)} y_{ij}(a_{ij} = a, \mathbf{a}_{i,-j}) \Pr(\mathbf{A}_{i,-j} = a_{i,-j} | A_{ij} = a)$$

Group average potential outcome

$$\bar{y}_i(a, \alpha) = \frac{1}{n_i} \sum_{j=1}^{n_i} \bar{y}_{ij}(a, \alpha)$$

Population average potential outcome

$$\bar{y}(a, \alpha) = \frac{1}{N} \sum_{i=1}^{N} \bar{y}_i(a, \alpha)$$

Average Potential Outcomes: Example

- Suppose that group 1 has the following potential outcomes $y_{1j}(\mathbf{x}_1)$

j	000	001	010	100	011	101	110	111
1	1	2	3	4	5	6	7	8
2	9	10	11	12	13	14	15	16
3	17	18	19	20	21	22	23	24

- Suppose completely randomized individual treatment assignment with $K_1 = 2$ for α_1 and $K_1 = 1$ for α_0

$$\bar{y}_1(0, \alpha_1) = ?$$
$$\bar{y}_1(0, \alpha_0) = ?$$

Average Potential Outcomes: Example

- Suppose that group 1 has the following potential outcomes $y_{1j}(\mathbf{x}_1)$

j	000	001	010	100	011	101	110	111
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3	17	18	19	20	21	22	23	24

- Suppose completely randomized individual treatment assignment with $K_1 = 2$ for α_1 and $K_1 = 1$ for α_0

$$\bar{y}_1(0, \alpha_1) = \frac{5 + 14 + 23}{3} = 14$$

$$\bar{y}_1(0, \alpha_0) = \frac{(2+3)/2 + (10+12)/2 + (19+20)/2}{3} = 11$$

Average Potential Outcomes

Marginal individual average potential outcomes

$$\bar{y}_{ij}(\alpha) = \sum_{a_i \in A(n_i)} y_{ij}(\alpha_1)\,\Pr(\mathbf{A}_i = \mathbf{a}_i)$$

Marginal group and population average potential outcomes

$$\bar{y}_i(\alpha) = \frac{1}{n_i}\sum_{j=1}^{n_i} \bar{y}_{ij}(\alpha)$$

$$\bar{y}(\alpha) = \frac{1}{N}\sum_{i=1}^{N} \bar{y}_i(\alpha)$$

Causal Estimands: Direct Effects

Individual direct causal effect of treatment 0 compared to treatment 1 for the individual j in group i by

$$CE_{ij}^{D}(\alpha) = y_{ij}(a_{ij} = 1, \alpha) - y_{ij}(a_{ij} = 0, \alpha)$$

Individual average direct causal effect

$$\overline{CE}_{ij}^{D}(\alpha) = \bar{y}_{ij}(1, \alpha) - \bar{y}_{ij}(0, \alpha)$$

Group average direct causal effect

$$\overline{CE}_{i}^{D}(\alpha) = \bar{y}_{i}(1, \alpha) - \bar{y}_{i}(0, \alpha)$$

Population average direct causal effect

$$\overline{CE}^{D}(\alpha) = \bar{y}(1, \alpha) - \bar{y}(0, \alpha)$$

Q: Which of the quantities above can never be identified in the observed data (A, Y)?

Causal Estimands: Indirect Effects

Individual indirect causal effect of treatment programs α_1 compared with α_1 on individual j in group i by

$$CE^I_{ij}(\alpha_1, \alpha_0) = y_{ij}(\alpha_1, a_{ij} = 0) - y_{ij}(\alpha_0, a_{ij} = 0)$$

Individual average indirect causal effect

$$\overline{CE}^I_{ij}(\alpha_1, \alpha_0) = \bar{y}_{ij}(0, \alpha_1) - \bar{y}_{ij}(0, \alpha_0)$$

Group average indirect causal effect

$$\overline{CE}^I_{i}(\alpha_1, \alpha_0) = \bar{y}_{i}(0, \alpha_1) - \bar{y}_{i}(0, \alpha_1)$$

Population average indirect causal effect

$$\overline{CE}^I(\alpha_1, \alpha_0) = \bar{y}(0, \alpha_1) - \bar{y}(0, \alpha_0)$$

Causal Estimands: Total and Overall Effects

Population average total causal effect

$$\overline{CE}^{T}(\alpha_1, \alpha_0) = \bar{y}(1, \alpha_1) - \bar{y}(0, \alpha_0)$$

Population average overall causal effect

$$\overline{CE}^{O}(\alpha_1, \alpha_0) = \bar{y}(\alpha_1) - \bar{y}(\alpha_0)$$

Causal Estimands: Remarks

- Total = direct + indirect
- Estimands in general depend on treatment allocation strategy
- Under no interference

$$y_{ij}(\mathbf{a}_i) = y_{ij}(\mathbf{a}'_i) \text{ for all } \mathbf{a}_i, \mathbf{a}'_i \text{ such that } a_{ij} = a'_{ij}$$

  - Indirect causal effects are zero
  - Total causal effect equals direct causal effect
  - Causal effects not dependent on the treatment strategies

Estimators

Assumption 1: Completely randomized assignment strategy
For $a = 0, 1$:

$$\widehat{Y}_j(a, \alpha) = \frac{\sum_j Y_{ij} I(A_{ij} = a)}{\sum_j I(A_{ij} = a)} = \frac{1}{n_i} \sum_j \frac{Y_{ij} I(A_{ij} = a)}{\Pr[A_{ij} = a | S_i = s]}$$

Under assumption 1, $E[\widehat{Y}_i(a, \alpha) | S_i = s] = \bar{y}_i(a, \alpha)$.

Example Revisited

- Suppose that group 1 has the following potential outcomes $y_{1j}(\mathbf{x}_1)$

j	000	001	010	100	011	101	110	111
1	1	2	3	4	5	6	7	8
2	9	10	11	12	13	14	15	16
3	17	18	19	20	21	22	23	24

- Under Assumption 1, with $K_1 = 2$ for α_1 and $K_1 = 1$ for α_0

$$E\{\widehat{Y}_i(0, \alpha_1)|S_i = 1\} = ?$$
$$E\{\widehat{Y}_i(0, \alpha_0)|S_i = 0\} = ?$$

Example Revisited

- Suppose that group 1 has the following potential outcomes $y_{1j}(\mathbf{x}_1)$

j	000	001	010	100	011	101	110	111
1	1	2	3	4	5	6	7	8
2	9	10	11	12	13	14	15	16
3	17	18	19	20	21	22	23	24

- Under Assumption 1, with $K_1 = 2$ for α_1 and $K_1 = 1$ for α_0

$$E\{\hat{Y}_1(0,\alpha_1)|S_1 = 1\} = \frac{5+14+23}{3} = 14 = \bar{y}_1(0,\alpha_1)$$

$$E\{\hat{Y}_1(0,\alpha_0)|S_1 = 0\} = \frac{1}{3}\{\frac{2+10}{2} + \frac{3+19}{2} + \frac{12+20}{2}\} = 11 = \bar{y}_1(0,\alpha_0)$$

Estimators

- Complete randomization: Average over individuals then assignments $\bar{y}_j(a, \alpha)$ same as average over assignments, then individuals $E[\hat{Y}_j(a, \alpha)]$
- $\widehat{CE}^D(\alpha_1) = \widehat{Y}(1, \alpha_1) - \widehat{Y}(1, \alpha_1)$
- $\widehat{CE}^I(\alpha_1, \alpha_0) = \widehat{Y}(0, \alpha_1) - \widehat{Y}(0, \alpha_0)$
- $\widehat{CE}^T(\alpha_1, \alpha_0) = \widehat{Y}(1, \alpha_1) - \widehat{Y}(0, \alpha_0)$

Overall Estimators

- $\widehat{Y}_i(\alpha) = \frac{\sum_j Y_{ij}}{n_i}$
- $\widehat{Y}(\alpha) = \frac{\sum_i \hat{Y}_i(\alpha) I[S_i=1]}{\sum_i I[S_i=1]}$
- Under Assumption 1, $E\{\widehat{Y}(\alpha)\} = \bar{y}(\alpha)$
- Unbiased estimator: $\widehat{CE}^O(\alpha_1, \alpha_0) = \widehat{Y}(\alpha_1) - \widehat{Y}(\alpha_0)$

Variance (H&H, 2008)

- Unbiased estimators of the variance of the estimators does not exist without further assumptions
- Stratified Interference (SI): Only matters how many were treated in group or cluster, and does not matter who was treated
- For a given $a_{ij} = a$, individual j in group i has
  1 potential outcome assuming no interference
  n_i potential outcomes assuming stratified interference
  $2^{n_i - 1}$ potential outcomes under no assumptions
- Under SI, simple random sampling and two stage cluster sampling yield unbiased estimators of variance of $\hat{Y}_i(0, \alpha)$ and $\hat{Y}(0, \alpha_1)$
- Variance estimators are unbiased when effect is additive, positively biased otherwise

Illustrative Example

Two-stage randomized placebo-controlled vaccine trial based on data from Ali et al. (2005)

	α	Vaccine ($X_{ij} = 1$)		Placebo ($X_{ij} = 0$)	
		Total	Cases	Total	Cases
		$\sum_j X_{ij}$	$\sum_j X_{ij} Y_{ij}$	$\sum_j (1 - X_{ij})$	$\sum_j (1 - X_{ij}) Y_{ij}$
1	α_1	12541	16	12541	18
2	α_1	11513	26	11513	54
3	α_1	10772	17	25134	119
4	α_0	8883	22	20727	122
5	α_0	5627	15	13130	92

α_0 is the allocation strategy for the group that randomized 50% to the treatment.
α_1 is the allocation strategy for the group that randomized 30% to the treatment.

Estimates of population average effects per 1000 individuals per year

Effect	Parameter	Estimate	Estimated Variance
Direct	$\overline{CE}^D(\alpha_1)$	1.30	0.856
Direct	$\overline{CE}^D(\alpha_0)$	3.64	0.178
Indirect	$\overline{CE}^I(\alpha_1, \alpha_0)$	2.81	3.079
Total	$\overline{CE}^T(\alpha_1, \alpha_0)$	4.11	0.672
Overall	$\overline{CE}^O(\alpha_1, \alpha_0)$	2.37	1.430

Indirect: 50% vaccine coverage results in 2.8 fewer cholera cases per 1000 unvaccinated individuals per year compared to 30% vaccine coverage

Overall: 50% vaccine coverage results in 2.4 fewer cholera cases per 1000 individuals per year compared to 30% vaccine coverage

Observational Studies

- Methods in the presence of interference often rely on randomization and the assumption of partial interference (e.g. Sobel, 2006; Hong and Raudenbush, 2006; Rosenbaum, 2007; Hudgens and Halloran, 2008)
- Later work provides a solution to the problem of interference in randomized and nonrandomized designs with nonoverlapping clusters Tchetgen Tchetgen and VanderWeele, 2012; Liu and Hudgens, 2014; Buchanan et al, 2018)
- Suppose a two-stage randomization not employed, but instead we have an observational study
- Tchetgen Tchetgen and VanderWeele (2012) suggest IPW estimator where all observations from group i are weighted by the inverse of probability of the treatment assignment vector $\mathbf{A}_i$ given $\mathbf{X}_i$
- Essentially, standardizing to a counterfactual study with a Bernoulli allocation mechanism
- Alternative approaches to standardize to a study with correlation between the treatment assignment mechanisms (Barkley, et al 2020; Papadogeorgou et al. 2019)

Parameters (TT & V)

- $\pi(a_{k,-i};\alpha) = \Pr(A_{k,-i} = a_{k,-i}) = \prod_{j=1,j\neq i}^{n_k} \alpha^{a_{kj}}(1-\alpha)^{1-a_{kj}}$
- $\pi(a_k;\alpha) = \Pr(A_k = a_k) = \prod_{j=1}^{n_k} \alpha^{a_{kj}}(1-\alpha)^{1-a_{kj}}$.
- $\bar{y}_{ki}(a,\alpha) = \sum_{a_{k,-i}} y_{ki}(a_i = a, a_{k,-i})\pi(a_{k,-i};\alpha)$.
- Averaging over all individuals in each cluster, then over all clusters, we define the population average potential outcome as $\bar{y}(a,\alpha) = \sum_{k=1}^{m}\{\sum_{i=1}^{n_k} \bar{y}_{ki}(a,\alpha)/n_k\}/m$.
- Define the marginal average potential outcome for individual i under allocation strategy α by $\bar{y}_{ki}(\alpha) = \sum_{a_k} y_{ki}(a_k)\pi(a_k;\alpha)$.
- Averaging over individuals within each cluster, then over all clusters, define the population average potential outcome as $\bar{y}(\alpha) = \sum_{k=1}^{m}\{\sum_{i=1}^{n_k} \bar{y}_{ki}(a)/n_k\}/m$.
- Effects can be defined such as the spillover or indirect $\overline{IE}(\alpha,\alpha') = \bar{y}(0,\alpha) - \bar{y}(0,\alpha')$

Assumptions

- Representativeness of the unexposed for the treatment response had they been exposed and vice versa conditional on baseline covariates (i.e., conditional exchangeability at the cluster level)
  $\Pr(\mathbf{A}_k = \mathbf{a}_k | \mathbf{L}_k, \mathbf{Y}_k(\cdot)) = \Pr(\mathbf{A}_k = \mathbf{a}_k | \mathbf{L}_k)$
- Homogeneity of treatment effects despite any variations that may occur in practice, and no multiple versions of treatment
- No measurement error in any variable needed for valid analysis
- No interference between clusters.
- Cluster-level positivity assumption for the propensity score.
  $\Pr(\mathbf{A}_k = \mathbf{a}_k | \mathbf{L}_k) > 0$

Estimators

Cluster-level propensity score can be calculated by adjusting with individual level covariates among those in the cluster.

$$f_{A_k|X_k}(A_k|X_k;\theta_x,\theta_s) = \int \prod_{i=1}^{n_k} h_{ki}(b_k;\theta_x)^{A_{ki}}\{1 - h_{ki}(b_k;\theta_x)\}^{1-A_{ki}} f_b(b_k;\theta_s)db_k$$

where $h_{ki}(b_k;\theta_x) = Pr(a_{ki} = 1|X_{ki}, b_k, \theta_x) = logit^{-1}(X_{ki}\theta_x + b_k)$ is a propensity score for individual i in cluster k and $f_b(\cdot;\theta_s)$ is the density of cluster specific random effect $b_k \sim N(0,\theta_s)$.

Flexible model that can allow for non-null association between A_{ki} and $A_{ki'}$ given $\mathbf{L}_k$ for $i \neq i'$

IPW estimation: IPW estimator

IPW estimator for group-level potential outcome:

$$\widehat{Y}_k^{ipw}(a,\alpha) = \frac{\sum_{i=1}^{n_k} \pi_k(A_{k,-i};\alpha) I(A_{ki}=a) Y_{ki}}{n_k f_{A_k|X_k}(A_k|X_k;\hat{\theta})}$$

IPW estimator for population average potential outcome:

$$\widehat{Y}^{ipw}(a,\alpha) = \frac{1}{m}\sum_{k=1}^{K} \widehat{Y}_k^{ipw}(a,\alpha)$$

Marginal group-level potential outcome:

$$\widehat{Y}_k^{ipw}(\alpha) = \frac{\sum_{k=1}^{n_k} \pi_k(A_k;\alpha) Y_{ki}}{n_k f_{A_k|X_k}(A_i|X_i;\hat{\theta})}$$

IPW estimator for population average potential outcome:

$$\widehat{Y}^{ipw}(\alpha) = \frac{1}{m}\sum_{k=1}^{K} \widehat{Y}_k^{ipw}(\alpha)$$

Population-level IPW estimators

$$\widehat{DE}(\alpha) = \widehat{Y}^{ipw}(a=0;\alpha) - \widehat{Y}^{ipw}(a=1;\alpha)$$

$$\widehat{IE}(\alpha,\alpha') = \widehat{Y}^{ipw}(a=0;\alpha) - \widehat{Y}^{ipw}(a=0;\alpha')$$

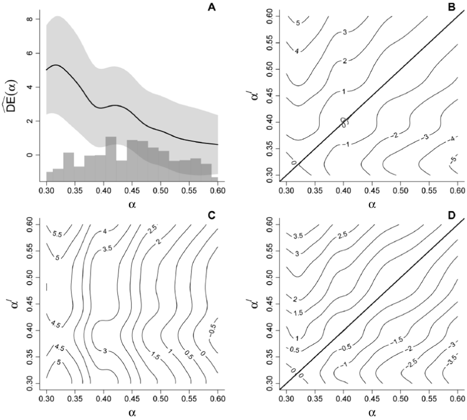
$$\widehat{TE}(\alpha,\alpha') = \widehat{Y}^{ipw}(a=0;\alpha) - \widehat{Y}^{ipw}(a=1;\alpha')$$

$$\widehat{OE}(\alpha,\alpha') = \widehat{Y}^{ipw}(\alpha) - \widehat{Y}^{ipw}(\alpha')$$

coverage: $\alpha < \alpha'$

Illustrative Example (Perez-Heydrich et al, 2014)

Individually randomized placebo-controlled vaccine trial based on data from Clemens et al. (1988) with (A) direct; (B) indirect; (C) total; and (D) overall.



Networks and Causal Inference

- Each node (person) has an outcome, treatment and covariates (attributes)
- Nodes are connected through edges, which represent social, work, school, sexual, healthcare, drug use/injection drug use, etc. partnerships
- Estimands: peer effects, treatment effects, spillover/interference effects, effects of network interventions
- Challenges:
  1. How to define and identify causal effects in a network-based study
  2. How to quantify uncertainty with complex network dependence

Non-randomized Interventions

- In many studies, the intervention or treatment is not randomized
- There may be confounding at either the individual, network-level or both
- Also face issues of network dependence and homophily
- Complex dependencies between observations
- Current methods employ
  - A generalized propensity score (Forastiere, 2021) or a Bayesian generalized propensity score (Forastiere, 2022) that account for individual and neighborhood covariates
  - Generalized propensity scores that do not assume stratified interference (Lee, 2023)
  - Doubly-robust estimator with exposure mapping (McNealis, 2024)
  - Targeted maximum likelihood estimation (TMLE) (Sofrygin, 2015; Van der Laan, 2014; Ogburn 2022)

Statistical Dependence in Networks

- Latent variables (i.e., homophily) lead to similar outcomes among close contacts
- Networks often observed at a single time point, so difficult to disentangle homophily from an effect
- Why is this a problem?
  - We cannot assume independence (i.e., cannot assume independent and identically distributed (iid))
  - Central limit theorem may not hold
  - Standard error estimates and confidence intervals will be anti-conservative!
- Network dependence (e.g., autocorrelation) is another threat to validity (particularly for single site studies) that can create bias (different from confounding and homophily!) (Lee, et al. 2023)

Possible Solutions for Dependence in Networks

THINK BIG WE DO℠

- Create conditionally independent units; analyze with standard models, but conditional on information barriers (Ogburn and Vanderweele, 2017)
- Extension of influence function from iid setting with interference set (van der Laan, 2014), social network setting with contagion and homophily (Ogburn, et al., 2017)
- Semiparametric inference with direct transmission and latent dependence (Ogburn et al., 2022)
- Nearest neighbor approach: Potential outcomes of any individual only depends their own exposure and on exposures of their nearest neighbors (or two-step neighbors) (Lee, et al, 2021; Forastiere, et al., 2020)
- Subsampling: Implementation and conditions may not be applicable to networks (e.g., bootstrap)
- K-dependence: $Cov(W_i, W_j) = \sigma_k$, where $k = ||i, j||$ and estimate using a plug-in estimator

Methodology

Recent methodological developments allow for interference sets within components, defined unique to each person

Nearest Neighbor IPW estimators

When partial interference assumption does not hold, we can use two inverse probability weighted (IPW) estimators where the interference set is the individual's *nearest neighbors* within the network.

1. IPW1 is a novel application for a sociometric network-based study setting (Liu et al., 2016).
2. IPW2 uses a generalized propensity score (Forastiere et al., 2016) with a weighted estimator instead of a stratified estimator.

Motivating Study: Transmission Reduction Intervention Project (TRIP)

- Sociometric network-based study of PWID and their contacts in Athens, Greece from 2013 to 2015.
- Intervention: **Community alerts**
  Outcome: **HIV risk behavior at 6-month follow up**

Baseline interview		Follow-up interview
• 277 participants • 29 alerted	six-month → time period	• 216 participants • 92 risk behavior

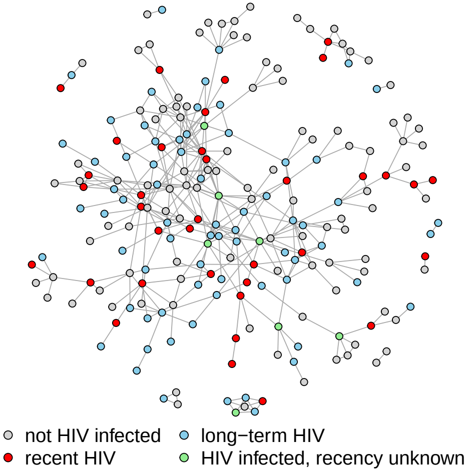


Figure 1: TRIP network with isolates removed. There are 216 participants and 362 links in the network.

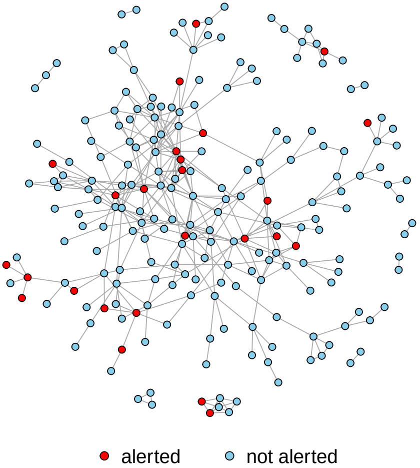


Figure 2: TRIP network with isolates removed. There are 216 participants and 362 links in the network.

Notations (1)

Let $i = 1, 2, \cdots, n$ denote each participant in the study.

- A_i: the self-selected binary treatment/exposure of participant i
- Z_i: the vector of covariates for participant i
- $\mathcal{N}_i$: the set of participants that share a link with i
- d_i: $|\mathcal{N}_i|$, the degree of node i
- $\mathbf{A}_{\mathcal{N}_i}$: the vector of baseline exposures for participants in $\mathcal{N}_i$
- $\mathbf{Z}_{\mathcal{N}_i}$: the vector of baseline covariates for participants in $\mathcal{N}_i$
- G_k: components (connected subnetworks) with $k = 1, \ldots, m$
- S_i: $= \sum_{j \in \mathcal{N}_i} A_j$; number of exposed nearest neighbors of individual i.

Notations (2)

Let $i = 1, 2, \cdots, n$ denote each participant in the study.

$y_i(a_i, \mathbf{a}_{\mathcal{N}_i})$: potential outcome of individual i if they received intervention a_i and their nearest neighbors received the vector of interventions denoted by $a_{\mathcal{N}_i}$

Y_i: $= y_i(A_i, \mathbf{A}_{\mathcal{N}_i})$ observed outcome (by consistency)

Estimand (1)

Under allocation strategy α, the probability for neighbors of i is denoted by $\pi(\mathbf{a}_{\mathcal{N}_i};\alpha) = \alpha^{\sum \mathbf{a}_{\mathcal{N}_i}}(1-\alpha)^{d_i - \sum \mathbf{a}_{\mathcal{N}_i}}$ and the probability for individual i is $\pi(a_i;\alpha) = \alpha^{a_i}(1-\alpha)^{1-a_i}$. In addition, $\pi(a_i, \mathbf{a}_{\mathcal{N}_i};\alpha) = \pi(\mathbf{a}_{\mathcal{N}_i};\alpha)\pi(a_i;\alpha)$

The population average potential outcome is defined by

$$\bar{y}(a,\alpha) = \frac{1}{n}\sum_{i=1}^{n}\sum_{\mathbf{a}_{\mathcal{N}_i}} y_i(a_i = a, \mathbf{a}_{\mathcal{N}_i})\pi(\mathbf{a}_{\mathcal{N}_i};\alpha)$$

and the marginal population average potential outcome is

$$\bar{y}(\alpha) = \frac{1}{n}\sum_{i=1}^{n}\sum_{a_i, \mathbf{a}_{\mathcal{N}_i}} y_i(a_i, \mathbf{a}_{\mathcal{N}_i})\pi(a_i, \mathbf{a}_{\mathcal{N}_i};\alpha)$$

Estimands (2)

Under allocation strategy α, the direct effect is

$$\overline{DE}(\alpha) = \bar{y}(1, \alpha) - \bar{y}(0, \alpha).$$

The spillover or indirect effect under allocation strategy $\alpha = (\alpha_1, \alpha_0)$ is

$$\overline{IE}(\alpha) = \bar{y}(0, \alpha_1) - \bar{y}(0, \alpha_0).$$

The composite or total effect is

$$\overline{TE}(\alpha) = \bar{y}(1, \alpha_1) - \bar{y}(0, \alpha_0).$$

The overall effect is

$$\overline{OE}(\alpha) = \bar{y}(\alpha_1) - \bar{y}(\alpha_0).$$

Assumptions

- Nearest neighbor interference: The potential outcomes only depends on the individual and their nearest neighbors $y_i(a_i, \mathbf{a}_{\mathcal{N}_i})$.
- Conditional exchangeability for participants:

$$\Pr(A_i = a_i | z_i, \mathbf{z}_{\mathcal{N}_i}, y_1(\cdot), \ldots, y_n(\cdot)) = \Pr(A_i = a_i | Z_i = z_i)$$

- Conditional exchangeability for neighbors:

$$\begin{aligned}&\Pr(A_i = a_i, \mathbf{A}_{\mathcal{N}_i} = \mathbf{a}_{\mathcal{N}_i} | z_i, \mathbf{z}_{\mathcal{N}_i}, y_1(\cdot), \ldots, y_n(\cdot)) \\ &= \Pr(A_i = a_i, \mathbf{A}_{\mathcal{N}_i} = \mathbf{a}_{\mathcal{N}_i} | z_i, \mathbf{z}_{\mathcal{N}_i})\end{aligned}$$

- Treatment variation irrelevance
- Positivity:

$$Pr(a_i, \mathbf{a}_{\mathcal{N}_i} | z_i, \mathbf{z}_{\mathcal{N}_i}) > 0 \text{ for all } a_i, \mathbf{a}_{\mathcal{N}_i}, z_i, \text{ and } \mathbf{z}_{\mathcal{N}_i}.$$

- Stratified interference assumption
- Reducible propensity score assumption

Nearest Neighbor IPW1 Estimator

The population average IPW1 estimator for $\bar{y}(a, \alpha)$ is defined as

$$\widehat{Y}^{IPW_1}(a, \alpha) = \frac{1}{n} \sum_{i=1}^{n} \frac{y_i(A_i, A_{\mathcal{N}_i}) I(A_i = a) \pi(A_{\mathcal{N}_i}; \alpha)}{f(A_i, A_{\mathcal{N}_i} | Z_i, \mathbf{Z}_{\mathcal{N}_i})}.$$

The marginal IPW1 estimator for $\bar{y}(\alpha)$ is defined as

$$\widehat{Y}^{IPW_1}(\alpha) = \frac{1}{n} \sum_{i=1}^{n} \frac{y_i(A_i, A_{\mathcal{N}_i}) \pi(A_i, A_{\mathcal{N}_i}; \alpha)}{f(A_i, A_{\mathcal{N}_i} | Z_i, \mathbf{Z}_{\mathcal{N}_i})}.$$

The neighborhood-level propensity score is defined as

$$f(A_i, A_{\mathcal{N}_i} | Z_i, \mathbf{Z}_{\mathcal{N}_i}) = \int \prod_{j \in \mathcal{N}_i^*} p_j^{A_j} (1 - p_j)^{1 - A_j} f(b_{\mathcal{N}_i^*}, 0, \theta_z) db_{\mathcal{N}_i^*}$$

where $\mathcal{N}_i^* = \mathcal{N}_i \cup \{i\}$ and $p_j = \text{logit}^{-1}(Z_i \cdot \theta_z + b_{\mathcal{N}_i^*})$ with $f(b_{\mathcal{N}_i^*}; 0, \psi) \sim N(0, \psi)$.

Nearest Neighbor IPW2 Estimator (1)

- Assumes stratified interference and reducible propensity score
- Potential outcomes of individual i depend on the total number of exposed neighbors, $s_i = \sum_{j \in \mathcal{N}_i} a_j$ (and $S_i = \sum_{j \in \mathcal{N}_i} A_j$):

$$y(a_i, \mathbf{a}_{\mathcal{N}_i}) = y(a_i, s_i)$$

- $f_2(A_i, S_i | Z_i, \mathbf{Z}_{\mathcal{N}_i}) = f_{21}(S_i | A_i, \mathbf{Z}_{\mathcal{N}_i}) f_{22}(A_i | Z_i) = C_{S_i}^{d_i} p_{1,i}^{S_i} (1 - p_{1,i})^{d_i - S_i} \cdot p_{2,i}^{A_i} (1 - p_{2,i})^{1 - A_i}$
- $p_{1,i} = Pr(S_i = s_i | A_i, \mathbf{Z}_{\mathcal{N}_i}) = \text{logit}^{-1}(A_i \beta + h(\mathbf{Z}_{\mathcal{N}_i}) \cdot \delta')$, and $p_{2,i} = Pr(A_i = 1 | Z_i) = \text{logit}^{-1}(Z_i \cdot \lambda)$

Nearest Neighbor IPW2 Estimator (2)

The population average IPW2 estimator for $\bar{y}(a, \alpha)$ is defined as

$$\widehat{Y}^{IPW_2}(a; \alpha) = \frac{1}{n} \sum_{i=1}^{n} \frac{y_i(A_i, S_i) I(A_i = a) \pi(S_i; \alpha)}{f_2(A_i, S_i | Z_i, \mathbf{Z}_{\mathcal{N}_i})}, \tag{1}$$

and marginal IPW2 estimator for $\bar{y}(\alpha)$ is defined as

$$\widehat{Y}^{IPW_2}(\alpha) = \frac{1}{n} \sum_{i=1}^{n} \frac{y_i(A_i, S_i) \pi(A_i, S_i; \alpha)}{f_2(A_i, S_i | Z_i, \mathbf{Z}_{\mathcal{N}_i})}, \tag{2}$$

with $\pi(S_i; \alpha) = C_{S_i}^{d_i} \alpha^{S_i} (1 - \alpha)^{d_i - S_i}$ and
$\pi(A_i, S_i; \alpha) = \pi(S_i; \alpha) \pi(A_i; \alpha)$.

G-computation estimator

Using a set-up similar to IPW2, define the outcome regression

$$\begin{aligned} Q(a_i, \mathbf{a}_{\mathcal{N}_i}, Z_i, \mathbf{Z}_{\mathcal{N}_i}) &= E(Y_i \mid A_i = a_i,\ \mathbf{A}_{\mathcal{N}_i} = \mathbf{a}_{\mathcal{N}_i},\ Z_i,\ \mathbf{Z}_{\mathcal{N}_i}) \\ &= E(Y_i \mid A_i = a_i,\ S_i = s_i,\ Z_i,\ \mathbf{Z}_{\mathcal{N}_i}) \\ &= Q(a_i, s_i, Z_i, \mathbf{Z}_{\mathcal{N}_i}) \end{aligned}$$

The g-computation estimator is

$$\hat{\bar{y}}^{\,\mathrm{gcomp}}(a, \alpha) = \frac{1}{n} \sum_{i=1}^{n} \sum_{s_i=0}^{d_i} \hat{Q}(a_i = a, s_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\, \pi(s_i; \alpha),$$

and

$$\hat{\bar{y}}^{\,\mathrm{gcomp}}(\alpha) = \frac{1}{n} \sum_{i=1}^{n} \sum_{a_i, s_i} \hat{Q}(a_i, s_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\, \pi(a_i, s_i; \alpha).$$

Why this augmented IPW form?

Start with g-computation:

$$\sum_{s_i} Q(a, s_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\, \pi(s_i; \alpha)$$

But this relies entirely on the model Q_i. Add a correction using observed data:

$$Y_i - Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})$$

Weight it to target the policy:

$$\frac{I(A_i = a)\, \pi(S_i; \alpha)}{f_{2i}(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})} \Big\{ Y_i - Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i}) \Big\}$$

Result:

$$\text{AIPW} = \text{plug-in mean} + \text{weighted residual}$$

- plug-in term: low variance, model-based
- residual term: corrects bias using observed data

Augmented IPW2 (AIPW2) estimator

The estimator is

$$\hat{\bar{y}}^{\,\mathrm{DR}}(a,\alpha) = \frac{1}{n}\sum_{i=1}^{n}\Bigg[\frac{I(A_i = a)\,\pi(S_i;\alpha)}{\hat{f}_2(A_i, S_i, \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})}\Big\{Y_i - \hat{Q}(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\Big\}$$
$$+ \sum_{s_i}\hat{Q}(a_i, s_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\,\pi(s_i;\alpha)\Bigg].$$

$$\hat{\bar{y}}^{\,\mathrm{DR}}(\alpha) = \frac{1}{n}\sum_{i=1}^{n}\Bigg[\frac{\pi(A_i, S_i;\alpha)}{\hat{f}_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})}\Big\{Y_i - \hat{Q}(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\Big\}$$
$$+ \sum_{a_i, s_i}\hat{Q}(a_i, s_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\,\pi(a_i, s_i;\alpha)\Bigg].$$

Property: Consistent if either Q is correctly specified or f_2 is correctly specified.

Why is the AIPW estimator doubly robust?

For fixed a and α, define

$$\psi(a,\alpha) = \bar{y}(a,\alpha) = \frac{1}{n}\sum_{i=1}^{n}\sum_{s_i=0}^{d_i} y_i(a,s_i)\pi(s_i;\alpha).$$

The probability limit for AIPW2 is

$$E\left[\frac{I(A_i=a)\pi(S_i;\alpha)}{f_2(A_i,S_i \mid Z_i,\mathbf{Z}_{\mathcal{N}_i})}\left\{Y_i - Q(A_i,S_i,Z_i,\mathbf{Z}_{\mathcal{N}_i})\right\} + \sum_{s_i} Q(a,s_i,Z_i,\mathbf{Z}_{\mathcal{N}_i})\pi(s_i;\alpha)\right].$$

So to prove double robustness, it is enough to show the first term has mean 0 if either Q or f_2 is correct.

Scenario 1: Outcome model is correct

Let

$$R = E\left[\frac{I(A_i = a)\pi(S_i;\alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})}\Big\{Y_i - Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\Big\}\right].$$

If Q_i is correctly specified, then

$$Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i}) = E\big(Y_i \mid A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i}\big).$$

Therefore

$$E\Big[Y_i - Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i}) \mid A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i}\Big] = 0.$$

Taking iterated expectations,

$$R = E\left[E\left\{\frac{I(A_i = a)\pi(S_i;\alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})}\Big(Y_i - Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\Big) \,\middle|\, A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i}\right\}\right] = 0.$$

Hence

$$E\{\hat{\psi}^{\mathrm{DR}}(a,\alpha)\} = E\left[\sum_{s_i} Q(a, s_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\pi(s_i;\alpha)\right] = \psi(a,\alpha).$$

Scenario 2: Propensity score model is correct

Now suppose $f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})$ is correctly specified, but Q may be misspecified. Consider the first term:

$$R = E\left[\frac{I(A_i = a)\pi(S_i;\alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})} Y_i\right].$$

Using iterated expectations,

$$R = E\left[\sum_{s_i} E\left\{\frac{I(A_i = a)\pi(S_i;\alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})} Y_i \,\middle|\, Z_i, \mathbf{Z}_{\mathcal{N}_i}, S_i = s_i\right\}\right].$$

Since the denominator is the true conditional law,

$$E\left[\frac{I(A_i = a)\pi(S_i;\alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})} Y_i \,\middle|\, Z_i, \mathbf{Z}_{\mathcal{N}_i}\right] = \sum_{s_i} \pi(s_i;\alpha)\, E\{Y_i(a, s_i) \mid Z_i, \mathbf{Z}_{\mathcal{N}_i}\}.$$

So

$$E\left[\frac{I(A_i = a)\pi(S_i;\alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})} Y_i\right] = \psi(a,\alpha).$$

Subtracting and adding the same weighted term with Q_i, the augmentation cancels, so

$$E\{\hat{\psi}^{\mathrm{DR}}(a,\alpha)\} = \psi(a,\alpha)$$

even if Q_i is misspecified.

Conclusion: double robustness

If $Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})$ is correct, then the residual term has mean zero:

$$E\left[\frac{I(A_i = a)\pi(S_i; \alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})}\left\{Y_i - Q(A_i, S_i, Z_i, \mathbf{Z}_{\mathcal{N}_i})\right\}\right] = 0.$$

If $f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})$ is correct, then the weighted term identifies the target:

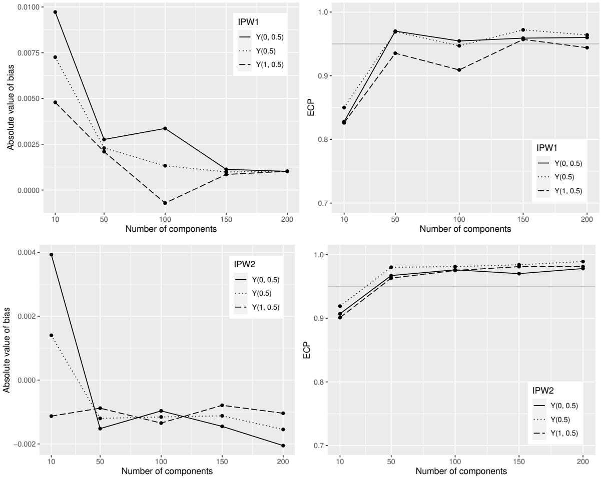
$$E\left[\frac{I(A_i = a)\pi(S_i; \alpha)}{f_2(A_i, S_i \mid Z_i, \mathbf{Z}_{\mathcal{N}_i})} Y_i\right] = \psi(a, \alpha).$$

Therefore, $\hat{\bar{y}}^{\,\mathrm{DR}}(a, \alpha)$ is consistent if either: the outcome model Q is correct, or the denominator model f_2 is correct. The same argument applies to $\hat{\bar{y}}^{\,\mathrm{DR}}(\alpha)$.

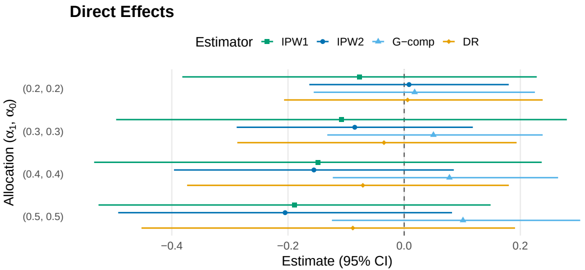
Variance Estimators

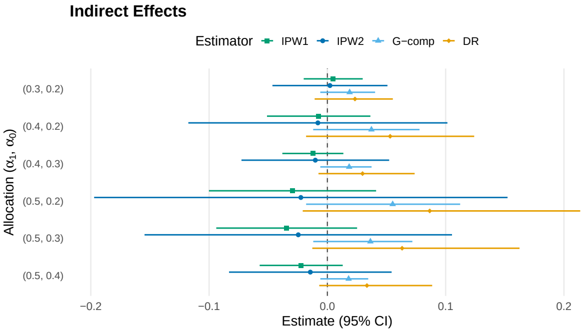
- Using M-estimation, we derived closed-form variance estimators for IPW1 and IPW2 that allow for overlapping interference sets
- Assume that the observed network can be expressed as the union of connected subnetworks, components of the network $\{G_1, G_2, \cdots, G_m\}$.
- Assume that the m components are a random sample from the infinite super-population of groups and the size of each component is bounded.
- We use m independent components (i.e., subnetworks), while we preserve the underlying connections comprising the network structure within each component, and assume $m \to \infty$.
- Can lead to more efficient variance estimator than assuming partial interference by components (Lee, et al. AOAS, 2023).
- For g-computation and doubly-robust estimators, we performed bootstrap resampling with sampling unit $\mathcal{N}_i^*$

Simulation Results for IPW1 and IPW2

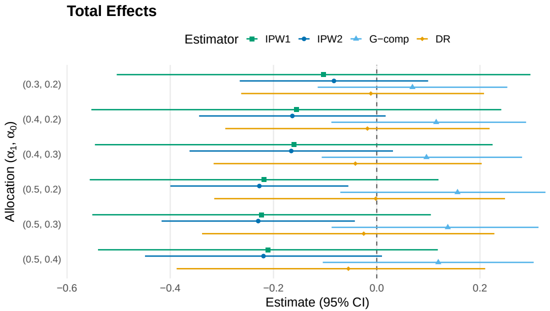


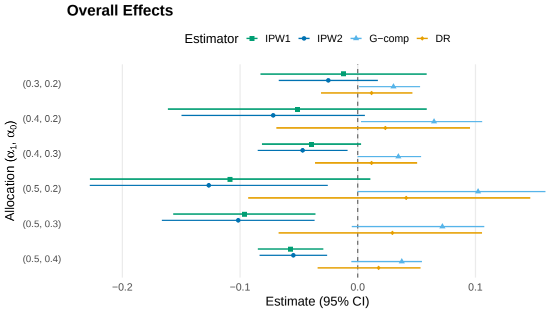
Alerts and HIV Risk Behavior in TRIP





Alerts and HIV Risk Behavior in TRIP





IPW2 Propensity Score Balance in TRIP

	Reduced model	Full model	Chisq	Pr(>Chisq)
1	date ~ ind_ps	date ~ alert_1 + ind_ps	0.1021	0.7493
2	posneg.x ~ ind_ps	posneg.x ~ alert_1 + ind_ps	0	0.9994
3	share.x ~ ind_ps	share.x ~ alert_1 + ind_ps	0.0022	0.9627
4	education ~ ind_ps	education ~ alert_1 + ind_ps	1e-04	0.9922
5	employ ~ ind_ps	employ ~ alert_1 + ind_ps	0	0.9977
6	avg_date ~ group_ps	avg_date ~ cbind(na_a, notna_a) + group_ps	0.4033	0.8174
7	avg_posneg.x ~ group_ps	avg_posneg.x ~ cbind(na_a, notna_a) + group_ps	0.2544	0.8806
8	avg_share.x ~ group_ps	avg_share.x ~ cbind(na_a, notna_a) + group_ps	1.6424	0.4399
9	avg_edu ~ group_ps	avg_edu ~ cbind(na_a, notna_a) + group_ps	0.5717	0.7514
10	avg_employ ~ group_ps	avg_employ ~ cbind(na_a, notna_a) + group_ps	2.8133	0.245

Considerations and Future Directions

- IPW1 normality of random effects (Tchetgen Tchetgen and Coull, 2006).
- Individual vs. cluster weighted estimands (Kahan et al. 2023, 2024; Basse and Feller, 2018)
- Policy-relevant estimands (Loomba and Eckles, 2025)
- Possibly underpowered to detect spillover effects (Baird et al. 2018; Jiang et al. 2023)
- G-null paradox with neighbor interference (Kilpatrick et al. 2024)
- More flexible estimation approaches, like TMLE in networks (van der Laan, 2014)

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https://github.com/uri-ncipher
https://web.uri.edu/ncipher/

Discussion Questions

- Which settings would you expect there to be interference? Which settings would you find the assumption of no interference plausible?
- Do you have any suggestions on how to disentangle homophily from a causal effect in a network?
- Can you think of any other ways to create (conditional) independence in a network?
- What other sources of bias should be addressed (e.g., selection bias)?

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Github R Code and TRIP Data

R functions available to assess spillover in networks.

https://github.com/uri-ncipher/Nearest-Neighbor-estimators
https://github.com/osofr/simcausal
https://github.com/vanessamcnealis/Network-Interference-estimators

Transmission Reduction Intervention Project, Athens site data: Nikolopoulos, Georgios K., Friedman, Samuel R., and Buchanan, Ashley L. Transmission Reduction Intervention Project (TRIP), Athens, Greece Site, 2013-2015. Inter-university Consortium for Political and Social Research [distributor], 2024-12-09.